

Project Workflow

1. Surveillance drones equipped with high-resolution cameras fly over agricultural fields to continuously capture real-time images of crops while sensors collect environmental data.
2. The captured high-quality images and sensor data are instantly transmitted to the central cloud infrastructure using ultra-fast 5G connectivity to ensure low-latency processing.
3. Raw images stored in the cloud undergo pre-processing to remove noise and enhance contrast, followed by image segmentation to isolate specific plant leaves for analysis.
4. The pre-processed data is fed into the EfficientNetV2B0 deep learning model, which extracts features and classifies each plant as either healthy or diseased using softmax activation.
5. If a plant is classified as diseased, the system flags the specific instance and records its exact GPS coordinates to pinpoint the location for treatment.
6. A smart spraying mechanism navigates to the flagged coordinates and applies pesticides only to the infected plants, minimizing chemical waste and reducing environmental impact.